

Weigh, Dim and Route a Parcel SOP

One episode handles one parcel. It starts with one parcel at the in spot, an empty scale, a bare measure mat, a blank log card and three sticker pads. It ends with the parcel standing on the measure mat, its weight and its three sides written on the card, one band box marked, and one routing sticker flat on its top face, with every tool back on its spot.

The order never changes: weigh the parcel, measure its three sides, write the four numbers on the card, read the band off the chart and mark it, put the routing sticker on. Nothing is written before the number it comes from is on the display or on the bar, and no sticker is peeled before the band box is marked.

The gripper does all the work. It takes the parcel from the in spot, sets it on the scale, moves it to the mat, works the bar, works the pen, peels the sticker and presses it flat. Nothing has to be held still, because nothing moves on its own: the scale, the mat, the card and the three sticker pads all grip the table, and a parcel standing on the mat stands there on its own.

Only one thing is held at a time. Nothing is passed from one grip to another, no grip is changed while an object is off the table, and nothing is dragged or slid along the table. Every pick lifts straight up off the surface before it moves anywhere.

The scale, the bar and the chart do the deciding, not the eye. A parcel is small, medium or large only because the number on the display and the number on the bar meet in that box on the chart. No parcel is weighed by hand, no parcel is measured by eye, and no two parcels are held up against each other.

The card is written in the plainest words the job allows. A parcel is not logged as oversize or out of gauge: it is marked SMALL, MEDIUM or LARGE, and the numbers it came from are written beside it. Anyone who can read those three words can read the card.

Each tool goes back to its own spot before the next step starts. The counts are fixed: one parcel, one weighing, three measurements, four numbers, one X, one sticker and one press of 2 seconds. There is no second weighing of the same parcel, no second measurement of the same side and no second sticker. A number that is written wrong is left on the card, tagged, and the episode goes on. Nothing is ever rubbed out or tidied up on camera.

Setup

Finish both checklists before starting an episode.

Cell configuration

Environment camera: 90 mm.

cell_type: single_arm

Hardware checklist

1. Cameras are on and recording.
2. The environment camera frame holds the whole tabletop in one shot: the in spot, the scale with its pan and its display, the measure mat, the bar spot, the log card, the pen holder and all three sticker pads.
3. The scale display stays in frame and readable the whole time, so a reviewer can read the same number the card was written from.
4. The scale pan stays in frame the whole time, so a reviewer can see whether the pan is empty or has a parcel on it.
5. The measure mat stays in frame with the whole parcel and the whole bar on it, so a reviewer can read the mark the bar was read at.
6. The four number boxes, the three band boxes and the band chart at the head of the card stay in frame and readable the whole time.
7. The gripper is at home and open.
8. The tabletop is dry and holds nothing but the items in the Materials checklist.
9. The gripper reaches the in spot, the scale pan, the ZERO bar, the whole measure mat, the bar spot, the card, the pen holder and all three sticker pads without stretching to a joint limit or folding in on itself. Bring a spot forward or closer in until this is true.
10. The gripper comes down onto the scale pan straight from above, so a parcel can be set down without being tilted, swung in or slid across the pan.
11. The paths from the in spot to the scale pan, from the pan to the measure mat, from the bar spot to the mat, from the pen holder to the card, and from each sticker pad to the mat are clear. Carry one parcel and the bar along each path by hand during setup: each one must clear the rim of the pan and the stops on the mat. Widen the gaps until it does.
12. Nothing leans on the scale, touches the scale or rests against it. Even a cable lying across the scale changes what it reads.
13. The table does not rock. Push down on the tabletop beside the scale during setup: if the display moves off zero, steady the table or move the scale.

Materials checklist

1. Parcels stand at the in spot at the back left, enough for the session, one taken per episode. Each one is a closed box with flat faces, square corners, a flat top face, and no sticker and no writing on it.
2. Each parcel is light enough for the gripper to carry level, and small enough to sit whole on the scale pan with a clear gap all the way round.
3. The parcels are not all the same size or the same weight. Some fall in each of the three bands, so the chart is read for real in every episode.
4. Each parcel is stiff enough to keep its shape. A parcel that squashes under the gripper, or whose top face bows, is not used: the bar cannot be read against a face that moves.

5. One bench scale sits at the back right, its pan level and its display facing the camera. It is heavy and it grips the table.
6. The scale reads in whole grams and it reads to at least 5000 grams. The display is big enough to read on camera.
7. The scale keeps the last number on the display until the ZERO bar is pressed. Replace a scale whose display goes blank or jumps back to zero when the parcel is lifted off.
8. The scale has one flat ZERO bar on its front face, wide enough to press with a closed gripper, labelled ZERO facing the camera. It is the only control that is ever touched.
9. The scale pan is empty, clean and dry at the start of the episode. A crumb or a chip left in the pan is weight the parcel gets blamed for.
10. The scale is on and warmed up, and with nothing on the pan it reads zero and holds there for 5 seconds without drifting.
11. One measure mat lies flat at the middle of the table, straddling the midline, bare at the start of the episode. It is wider and deeper than the biggest parcel in the set, with room for the bar to lie beside the parcel on every side.
12. The mat has one back stop along its far edge and one side stop up its left edge, both raised, both square to each other, and both fixed to the mat. A parcel pushed against both sits square.
13. The mat grips the table. Push it by hand during setup: if it slides, turns or lifts, weigh it down or clamp it. Nothing pins the mat during the episode.
14. One measuring bar lies flat on the bar spot, to the right of the mat, with its end stop toward the back of the table. It is a stiff flat bar, longer than the longest side of the biggest parcel.
15. The bar carries whole centimetre marks along one edge, numbered from its end stop, printed big enough to read on camera. The marks face up when the bar lies on its spot.
16. The bar has one raised end stop at its zero end. The zero mark sits level with the inside face of that stop, so a reading starts at the stop and not before it.
17. One log card lies flat at the front centre on the midline. It is a wipe clean card and it starts the episode blank.
18. The card carries four number boxes stacked down its left side, labelled in full words down their left: WEIGHT, LENGTH, WIDTH, HEIGHT. Each box is big enough to hold one number with room to spare.
19. The card carries three band boxes in a line across its foot, labelled in full words: SMALL, MEDIUM, LARGE. Every box is big enough to hold one X with room to spare, and the boxes do not touch each other.
20. The card carries the band chart at its head, printed in full plain words and whole numbers: three weight rows, three length columns, and one band word printed in each of the nine boxes where a row and a column meet.
21. The words on the card are printed in full and in plain English. No code, no service name and no parcel number appears anywhere on it.

22. The card grips the table. Push it by hand during setup: if it slides, tape it down or put it in a holder.
23. One dry erase pen stands in its holder on the card's left edge, cap off, with enough ink to write four numbers and one X.
24. Three sticker pads lie flat in a line at the front right, left to right: the SMALL pad, the MEDIUM pad, then the LARGE pad. Each pad is labelled facing the camera and holds stickers of one band only.
25. Every sticker on a pad prints its own band word, SMALL, MEDIUM or LARGE, in full, big enough to read on camera. The three bands are clearly different in colour.
26. The corner of the next sticker on each pad is lifted off the pad, so the gripper has an edge to close on. A sticker lying flat with no lifted corner is not usable: the gripper cannot lift the corner and peel the sticker at the same time.
27. Each pad holds more than one sticker, so a peel that goes wrong can be followed by a clean one.
28. A clear gap of at least a gripper's width sits between the scale and the mat, between the mat and the bar spot, and between each pair of sticker pads, so nothing is knocked on the way past.

Workspace layout

Three rows: the parcels and the scale at the back, the measure mat in the middle with the bar beside it, the card and the sticker pads at the front. The parcel travels right to the scale, then forward onto the mat, and it stays there.

Back row, left to right:

1. **In spot**, back left. The waiting parcels stand here. It empties as the session runs.
2. **Scale**, back right, its pan level and empty at the start of the episode, its display facing the camera and its ZERO bar on the front face.

Middle row, left to right:

1. **Measure mat**, the middle of the table, straddling the midline, with its back stop along its far edge and its side stop up its left edge. Everything that happens to the parcel after the weighing happens here.
2. **Bar spot**, to the right of the mat, the measuring bar lying flat with its marks facing up and its end stop toward the back of the table.

Front row, left to right:

1. **Pen holder**, on the card's left edge.
2. **Log card**, front centre on the midline, its four number boxes, its three band boxes and its band chart facing the camera.
3. **Sticker pads**, front right, three pads in a line left to right: SMALL, MEDIUM, LARGE.

No carry passes over the scale once the parcel is off it, no carry passes over the card, and nothing travels to the far side of the table. Move the parcel back or forward on its own line first, then straight across.

Vocabulary

Parcel: one closed box. There is one in every episode, and it is the output of the episode.

In spot: the place at the back left the waiting parcels stand. One parcel is taken from it per episode.

Scale: the bench scale at the back right. It is the only thing that says what the parcel weighs.

Pan: the flat tray on top of the scale that the parcel is set on. Only the parcel sits on it, and nothing else is ever set on it.

Display: the window on the scale that shows the weight in whole grams. It faces the camera and it is never blocked while a number is being read.

ZERO bar: the flat bar on the front of the scale. Pressing it once sets the display back to zero. It is the only control that is ever touched.

Zeroed: the pan is empty and the display reads zero. The scale is zeroed at the start of every episode.

Steady reading: the number on the display has stopped changing and has held the same for 2 seconds with nothing touching the parcel, the pan, the scale or the table.

Held number: the scale keeps the steady reading on the display after the parcel is lifted off, until the ZERO bar is pressed again. The weight stays in view for the whole episode.

Measure mat: the flat mat in the middle of the table the parcel is measured on. The parcel stays on it from Step 2 to the end of the episode.

Mat stops: the back stop along the mat's far edge and the side stop up its left edge. A parcel pushed against both sits square.

Seated parcel: a parcel standing flat on the mat, touching both stops, with its longest side running left to right and its shortest side standing up, and not rocking.

Measuring bar: the stiff flat bar with whole centimetre marks along one edge and a raised end stop at its zero end. It is laid against the parcel and read, and it is never used to push the parcel.

End stop: the raised end of the bar. It sits against a mat stop or flat on the mat, and every reading starts there.

Mark: one whole centimetre line on the bar. A reading is one mark and it is read where a face of the parcel crosses the bar.

Length: the side of the parcel that runs left to right on the mat. It is the longest of the three sides.

Width: the side that runs front to back on the mat.

Height: the side that stands up from the mat to the top face.

Top face: the face of the parcel pointing at the ceiling once it is seated. The sticker goes in the middle of it.

Log card: the wipe clean card at the front centre. It starts the episode blank and ends it with four numbers and one X.

Number box: one of the four boxes down the left of the card, labelled WEIGHT, LENGTH, WIDTH or HEIGHT. It holds one number, written in full, and nothing else.

Band chart: the nine boxes printed at the head of the card, three weight rows across three length columns, with one band word printed in each box. It is read, never written on.

Band: the one word the chart gives a weight and a length: SMALL, MEDIUM or LARGE. Every parcel has exactly one band.

SMALL: the box where the weight row and the length column meet says SMALL. The parcel takes a SMALL sticker.

MEDIUM: the box where the weight row and the length column meet says MEDIUM. The parcel takes a MEDIUM sticker.

LARGE: the box where the weight row and the length column meet says LARGE. The parcel takes a LARGE sticker.

Band box: one of the three boxes across the foot of the card. Exactly one of them carries an X, and it is the one the chart says.

Marked box: a box on the card holding its mark, fully inside its lines and not touching the next box.

Pen: the dry erase pen in its holder on the card's left edge. It writes on the card and on nothing else.

Sticker pad: one of the three pads at the front right. Each pad holds stickers of one band only.

Lifted corner: the corner of the next sticker on a pad raised off the pad. It is the grasp: the gripper cannot lift a sticker that lies flat.

Routing sticker: one sticker printed with its band word. One goes on the parcel in every episode.

Flat sticker: a sticker pressed down with no lifted corner, no crease and no bubble under it.

Press: the closed gripper pushed onto the sticker for 2 seconds, from the middle of the sticker out to each of its four edges.

Tool home: the spot a tool goes back to and sits on before the next step starts: the bar spot, the pen holder. A tool is never left on the mat, on the card, on the scale or in the gripper between steps.

Release point: the gripper opens over the middle of the pan when the parcel is resting on it, over the mat when the parcel is being set down, over the top face when the sticker is being laid down, or over a tool home. Never over the card, over the scale or over open tabletop.

Stable placement: the object stands still after release, square on its surface, and does not rock, tip, slide or lean on anything.

Spill: a parcel, the bar, the pen or a sticker that ends up off the table or somewhere it does not belong. It is left where it falls and cleared in the reset, never on camera.

Quarter turn: the most any object is tilted in the gripper. Everything travels level, and nothing is ever turned over in the air.

Home: the resting pose, clear of the table with the gripper open.

Steps

Step 1: Weigh the parcel

Goal: the parcel standing on its own in the middle of the pan, let go of, and one steady number on the display.

The scale and the mat both grip the table, so nothing holds either of them while the parcel is moved. The parcel is set down and then let go of. It is never held while the number is being read, because a grip still on the parcel is weight the scale is reading too.

1.1 Zero the scale

Check that the pan is empty, clean and dry, with nothing leaning on the scale.

With the gripper closed, come down onto the ZERO bar straight from the front, press it once, and come away.

One press. Do not hold the bar down and do not press it twice.

Hold still and wait 2 seconds with nothing touching the scale or the table.

Nothing is brushed off the pan with the gripper. A pan with something in it is a reset job.

Check: the display reads zero and holds there. If it does not read zero, or it drifts off zero, end the episode and fix the scale during reset. Do not press the bar again to chase it.

1.2 Set the parcel on the pan

Close on the sides of one parcel at the in spot, square to it, so it hangs level.

Lift it straight up off the table until it is clear, carry it to the scale keeping it level and clear of the display, and bring it over the middle of the pan.

Lower it straight down slowly until it takes its own weight on the pan, then stop.

Do not drop it onto the pan, do not slide it across the pan and do not press it down.

Take one parcel per episode. Never lift two at once, and never drag a parcel across the table.

Check: the parcel sits whole inside the pan, square and level, with a clear gap all the way round it, and no part of it hangs over the rim. If it hangs tilted or overhangs the rim, lift it clear and set it down again. Make no more than one correction.

1.3 Let go and read the number

Open the gripper and take it right away from the parcel, the pan and the scale.

Hold still and wait 2 seconds with nothing touching the parcel, the scale or the table.

The number on the display has to stop changing and hold the same for those 2 seconds.

Read the steady number. Leave it on the display. It stays there until the next episode zeroes the scale.

Do not steady the parcel, do not tap the pan and do not press the ZERO bar with the parcel on it.

Check: the display is readable in frame, the gripper is clear of it, and the number has held still. If the number is still moving after 5 seconds, end the episode and let review flag it.

Step 2: Measure the three sides

Goal: the parcel seated square on the mat, and its length, width and height each read off the bar as one whole centimetre mark, with the bar back on its spot.

The parcel is measured where it stands. It is not lifted again between the three readings, so all three numbers come from the same seating.

2.1 Move the parcel onto the mat and seat it square

Close on the sides of the parcel square, lift it straight up off the pan until it is clear of the pan rim, and carry it forward onto the mat.

Set it down with its longest side running left to right and its shortest side standing up, then open the gripper and lift clear.

If the longest side is not running left to right, close on the parcel, lift it straight up, turn it a quarter turn, and set it down again.

With the gripper closed and empty, push the parcel back until its far face touches the back stop, then push it left until its left face touches the side stop.

Open the gripper and lift clear.

Check: the parcel stands flat on the mat, touches both stops, does not rock, and its longest side runs left to right with its shortest side standing up. If it rocks or sits off a stop, seat it again. Make no more than one correction.

2.2 Measure the length

Close on the bar at its far end, lift it straight up off the bar spot, and carry it to the mat.

Lay it flat on the mat in front of the parcel, running left to right, with its marks facing up.

Push it back until it runs along the front face of the parcel, and push it left until its end stop touches the side stop of the mat, so the zero mark lines up with the left face of the parcel.

Read the mark on the bar at the right face of the parcel. If the face falls between two marks, take the higher mark.

Keep the bar flat on the mat and do not lift the parcel to fit it.

Check: the bar lies flat, its end stop is against the side stop, and the right face of the parcel crosses a mark that is readable in frame. If the bar has moved the parcel off a stop, lay the bar down clear of the parcel, seat the parcel again, and lay the bar again. Make no more than one correction.

2.3 Measure the width

Lift the bar straight up off the mat, turn it a quarter turn so it runs front to back, and lay it flat on the mat to the right of the parcel.

Push it left until it runs along the right face of the parcel, and push it back until its end stop touches the back stop of the mat, so the zero mark lines up with the far face of the parcel.

Read the mark on the bar at the front face of the parcel. If the face falls between two marks, take the higher mark.

Check: the bar lies flat, its end stop is against the back stop, and the front face of the parcel crosses a mark that is readable in frame.

2.4 Measure the height

Lift the bar straight up off the mat, turn it a quarter turn so it stands upright, and bring it down against the front face of the parcel with its marks facing the camera.

Lower it until its end stop rests flat on the mat, and hold it upright and touching the front face.

Read the mark on the bar at the top face of the parcel. If the top face falls between two marks, take the higher mark.

Do not lean the bar over the parcel and do not rest it on the top face.

Check: the bar stands upright with its end stop on the mat, it touches the front face, and the top face of the parcel crosses a mark that is readable in frame. If the bar leans, stand it again and read again. Make no more than one correction.

2.5 Return the bar to its spot

Lift the bar clear of the parcel, carry it back to the bar spot, lay it flat with its marks facing up and its end stop toward the back of the table, and open the gripper.

Check: the bar lies fully on its spot, flat, not on the mat, not on the card and not against the parcel. The parcel still stands seated against both stops.

Step 3: Write the four numbers on the card

Goal: the weight and the three sides in their own boxes on the card, and the pen back in its holder.

The parcel stands on the mat and the weight is still on the display while the card is written, so every number being copied is still in view to be seen.

3.1 Take the pen

Close on the pen barrel, lift it straight up out of its holder, and bring the tip over the card.

Check: the pen hangs in the gripper with its tip down and its holder stands empty.

3.2 Write the four numbers

Write the number the display is showing in the WEIGHT box, in full, fully inside the box.

Write the length in the LENGTH box, the width in the WIDTH box and the height in the HEIGHT box, each in full and fully inside its own box.

Write them in that order: weight, length, width, height.

Copy each number exactly as it was read. Do not round it, shorten it or work anything out from it.

One number per box, written once. Nothing else is written on the card at this step.

Check: the card carries four numbers, one in each box, each inside its lines, the weight matches the display, and no number is in the wrong box. A wrong number is not rubbed out and not written over. Leave it and let review flag the episode.

3.3 Return the pen to its holder

Carry the pen back to its holder, lower it in until it stands on its own, release, and lift clear.

Check: the pen stands in its holder and nothing is left lying on the card.

Step 4: Read the band off the chart and mark it

Goal: one X in the band box the chart says, and the pen back in its holder.

The chart at the head of the card decides the band. It is read, never written on, and it is never worked out in the head.

4.1 Read the band

Find the weight row on the chart that holds the number in the WEIGHT box: under 500 grams, 500 to 2000 grams, or over 2000 grams.

Find the length column that holds the number in the LENGTH box: under 20 cm, 20 to 40 cm, or over 40 cm.

Read the band word printed in the box where that row and that column meet.

The chart reads the same three lines every time. Under 500 grams: SMALL under 20 cm, MEDIUM 20 to 40 cm, LARGE over 40 cm. From 500 to 2000 grams: MEDIUM under 20 cm, MEDIUM 20 to 40 cm, LARGE over 40 cm. Over 2000 grams: LARGE at every length.

Use the length and the weight only. The width and the height are written down but they do not choose the band.

Check: one band word has been read off one box of the chart, and that box is the one where the weight row and the length column meet.

4.2 Mark the band box

Close on the pen barrel, lift it straight up out of its holder, and bring the tip over the card.

Draw one X in the band box the chart says: SMALL, MEDIUM or LARGE.

One X, in one box. Draw it fully inside the box lines, not touching the box beside it.

The numbers say what the parcel is and the X says which sticker it takes. Nothing else is written on the card.

Check: the card carries one X, it is in the band the chart gave, and the other two band boxes are blank.

4.3 Return the pen to its holder

Carry the pen back to its holder, lower it in until it stands on its own, release, and lift clear.

Check: the pen stands in its holder and the card carries four numbers and one X.

Step 5: Put the routing sticker on the parcel

Goal: one sticker of the marked band lying flat and level in the middle of the top face, its word facing up and readable.

The parcel stays on the mat for this whole step. It is not lifted again, so the sticker goes onto a parcel that is standing on its own.

5.1 Peel one sticker

If the card is marked SMALL, go to the SMALL pad. If it is marked MEDIUM, go to the MEDIUM pad. If it is marked LARGE, go to the LARGE pad.

Close on the lifted corner of the next sticker on that pad.

Peel the sticker off in one move, up and toward the front of the table, keeping its word facing up the whole way.

Take one sticker per episode, from one pad only. Never reach across to another pad and never carry two stickers.

Check: one whole sticker hangs from the gripper, word facing up, and it came off the pad the X says. If the sticker tears, folds onto its own sticky side, or a second sticker comes away with it, lay it flat on the table clear of the pads, open the gripper, and peel the next sticker once. Make no more than one extra peel.

5.2 Lay it on the top face

Carry the sticker to the parcel and hold it 1 to 2 cm above the middle of the top face, still clear of it.

Turn it until its long edges run left to right, the same way as the length of the parcel, within 5 degrees, with its word the right way up for the camera.

Move it until it sits near the middle of the top face, with at least 2 cm of bare face past every side of it.

Lower it straight down until the whole sticky side is touching, then open the gripper and lift clear.

Make no more than two small moves to line it up. **The sticker is lined up before it touches, not after.**

Once the sticker is down it is not slid, nudged or turned. A sticker that landed crooked is left as it landed.

Check: the sticker lies on the top face, near the middle, clear of every edge, with its word facing up and readable in frame.

5.3 Press it flat

Close the gripper and press the sticker down for 2 seconds, starting at the middle and working out to each of its four edges.

Bring the gripper straight up off the sticker at the end of the press.

Do not press hard enough to move the parcel off its stops.

Check: the sticker lies flat with no lifted corner, no crease and no bubble under it, and the parcel still stands seated on the mat. If one corner lifts, press that corner down once more. Make no more than one correction.

Step 6: End the episode

Check the parcel: standing on the mat, touching both stops, with one sticker flat on its top face and its word readable.

Check the card: four numbers in the four boxes, one X in one band box, and the X matching the sticker on the parcel.

Check the display: still showing the weight that was written.

Check the tools: the bar flat on its spot, the pen standing in its holder.

Fix a failed check only with the fix written in its step. If no fix is written, or the fix would need an earlier step run again, end the episode and reset instead.

Go home with the gripper open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take the finished parcel off the mat, peel the sticker off it, and set the parcel aside for the next stage. A parcel is never used twice with a sticker still on it.
2. Wipe the card clean, so it starts the next episode blank with only its printed labels and its band chart on it.
3. Press the ZERO bar by hand and check that the empty pan reads zero and holds there for 5 seconds.
4. Lift the corner of the next sticker on each of the three pads off the pad, so the gripper has an edge to close on. Top up any pad holding fewer than two stickers.
5. Clear away any sticker, backing paper or scrap left on the table, and pick up anything that landed off the table.
6. Top up the in spot so it holds enough parcels for the rest of the session, each one closed, with a flat top face and no sticker and no writing on it.
7. Wipe any dust off the pan, the mat, the bar and the card, and dry them.
8. Lay the bar flat on the bar spot with its marks facing up and its end stop toward the back, and stand the pen in its holder with the cap off.
9. Check that the mat still grips the table and both its stops are square, the bar is not bent and its marks are still readable, and the pen still writes.
10. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

Visible cue: the parcel goes onto the mat before a steady number is on the display, the bar comes out before the parcel is seated, a number is written before it has been read, the band

box is marked before the four numbers are on the card, a sticker is peeled before the band box is marked, or a step is skipped.

SOP rule broken: Steps 1 to 5, weigh, measure, write, mark, sticker, in that order.

Coaching note: weigh it, measure it, write it, mark it, sticker it. Nothing is written before the number it comes from is in view.

Violation: Parcel held while it is weighed

Visible cue: the gripper is still closed on the parcel, or touching it, the pan, the scale or the table, while the number is being read.

SOP rule broken: Step 1.3, let go of the parcel and take the gripper right away before the number is read.

Coaching note: let go, back off, then read. A grip still on the parcel is weight the scale is reading too.

Violation: Scale not zeroed, or zeroed at the wrong time

Visible cue: the parcel goes onto the pan before the ZERO bar has been pressed and the display has read zero, the bar is pressed with the parcel on the pan, the bar is pressed more than once, or it is held down.

SOP rule broken: Step 1.1, one press on an empty pan, then the display reads zero before the parcel goes on.

Coaching note: empty pan, one press, wait for zero. The bar is never touched again once the parcel is on.

Violation: Weight read while the number is moving

Visible cue: the number is read and written while the display is still changing, or before it has held the same for 2 seconds.

SOP rule broken: Step 1.3, the number stops changing and holds the same for 2 seconds before it is read.

Coaching note: wait for the number to settle. A moving number is not a reading.

Violation: Parcel seated wrong on the mat

Visible cue: the parcel is off the back stop, off the side stop or off both, its longest side is not running left to right, its shortest side is not standing up, or it rocks when the gripper opens.

SOP rule broken: Step 2.1, the far face on the back stop, the left face on the side stop, longest side left to right, shortest side standing up.

Coaching note: push it back, then push it left, then let go. Every one of the three numbers comes from that seating.

Violation: Parcel moved between readings

Visible cue: the parcel is lifted, turned, slid or knocked off a stop at any point between the length reading and the height reading.

SOP rule broken: Step 2, all three sides are read from one seating and the parcel is not lifted again.

Coaching note: seat it once and leave it. A parcel that moved gives three numbers that do not belong together.

Violation: Bar laid or stood wrong

Visible cue: the end stop is not against the mat stop for the length or the width, the bar does not touch the face it is measuring, it lies at an angle instead of straight along the parcel, it leans while the height is read, or it rests on the top face of the parcel.

SOP rule broken: Steps 2.2, 2.3 and 2.4, the end stop sits against a mat stop or flat on the mat, and the bar runs straight along the face it is measuring.

Coaching note: stop against stop, bar against face, then read. A bar at an angle always reads long.

Violation: Bar used to push the parcel

Visible cue: the bar is pushed against the parcel hard enough to slide it, turn it or lift it off a stop, or the bar is used to shove the parcel back into its corner.

SOP rule broken: Step 2, the bar is laid against the parcel and read, never used to move it.

Coaching note: the closed gripper seats the parcel and the bar only measures it.

Violation: Measurement skipped or read off nothing

Visible cue: fewer than three readings are taken, the same side is measured twice, or a number is written for a side the bar was never laid against.

SOP rule broken: Step 2, three readings, one for the length, one for the width and one for the height.

Coaching note: three sides, three readings, in order. Nothing is measured by eye.

Violation: Number written wrong or in the wrong box

Visible cue: the weight on the card does not match the display, a side number does not match the mark the bar was read at, a number is in the wrong box, or a box is left blank.

SOP rule broken: Step 3.2, four numbers copied exactly, one in each of the four boxes.

Coaching note: copy what is in view, box by box, in order. Nothing is worked out on the way from the display to the card.

Violation: Writing rubbed out or written over

Visible cue: a number or an X is wiped off, scribbled out, or written over the top of another one.

SOP rule broken: Steps 3.2 and 4.2, a wrong mark is left on the card and review flags the episode.

Coaching note: leave it and go on. A cleaned up card hides what happened.

Violation: Band read wrong off the chart

Visible cue: the X sits in a band box that is not the one where the weight row and the length column meet, the width or the height is used to pick the band, or the band is chosen with no look at the chart.

SOP rule broken: Step 4.1, the band is the word printed where the weight row and the length column meet.

Coaching note: find the row, find the column, read the word. Only the weight and the length choose the band.

Violation: More than one band marked

Visible cue: two or three band boxes carry a mark, an X crosses into the box beside it, or no band box is marked at all.

SOP rule broken: Step 4.2, one X, in one box, fully inside its lines.

Coaching note: one X and one only, well inside its box.

Violation: Wrong sticker used

Visible cue: the sticker on the parcel does not print the same word as the marked band box, or the sticker is taken from a pad the X does not point to.

SOP rule broken: Step 5.1, the sticker comes from the pad the marked band box names.

Coaching note: read the X, then go to that pad. The word on the parcel and the word on the card always match.

Violation: Bad sticker used

Visible cue: a sticker that is torn, folded onto its own sticky side, or doubled with a second sticker is carried to the parcel and laid down on it.

SOP rule broken: Step 5.1, a bad sticker is laid aside and the next one is peeled.

Coaching note: a bad sticker is never used. Set it aside and peel a clean one.

Violation: Sticker placed wrong

Visible cue: the sticker lands off the top face, hangs over an edge, sits closer than 2 cm to any edge, sits more than 5 degrees out of square, or ends with its word upside down or facing the table.

SOP rule broken: Step 5.2, the sticker lies near the middle of the top face, square within 5 degrees, word facing up, clear of every edge by at least 2 cm.

Coaching note: square it and centre it while it is still above the face. Once it touches, it stays where it landed.

Violation: Sticker pushed into place

Visible cue: the sticker is slid, nudged or turned after it has touched the top face.

SOP rule broken: Step 5.2, once the sticker is down it is not moved.

Coaching note: nothing on this station is pushed into line. A crooked sticker is left crooked and review flags it.

Violation: Press skipped or short

Visible cue: the sticker is let go onto the parcel and the step ends with no closed gripper press, or the press is shorter than 2 seconds.

SOP rule broken: Step 5.3, press for 2 seconds from the middle out to each of the four edges.

Coaching note: hold the full 2 seconds. A quick tap lets the sticker lift later.

Violation: Tool not returned to its spot

Visible cue: the bar is left on the mat, on the card or against the parcel; the pen is left on the card or on the table; the gripper opens while the pen is still above its holder; or a tool is still in the gripper when the next step begins.

SOP rule broken: Steps 2.5, 3.3 and 4.3, each tool goes back to its own spot and sits flat there before the next step starts.

Coaching note: the return is part of the step, not a gap between steps. Lower until it touches, then let go.

Violation: Spill cleaned up

Visible cue: after a parcel, the bar, the pen or a sticker lands somewhere it should not, the gripper is used to sweep, push, wipe or pick it up.

SOP rule broken: Steps 1 to 5, a spill is left where it falls and is cleared in the reset.

Coaching note: cleanup moves are not part of the task. The spill is data, the cleanup is noise.

Violation: Bad pick, carry, or release

Visible cue: an object moves from one grip to another without being set down first, two objects are held at once, a grip is shifted or taken again while the object is off the table, the parcel or the bar or the pen slides along the table with no daylight under it, the parcel hangs tilted because it was taken off square, or an object is turned past a quarter turn or turned over in the air.

SOP rule broken: Steps 1 to 5, one object at a time, lifted straight up off the surface before any sideways move, gripped square, tilted no more than a quarter turn, and never passed between grips.

Coaching note: set it down to change the grip, and lift before the carry. A parcel that hangs tilted was taken off square.

Violation: Placement pushed, or correction limit passed

Visible cue: the parcel, the mat, the bar, the card or a sticker is shoved, nudged or slid into place instead of being lifted clear and placed again, or a failed check is tried again more often than its step allows. Seating the parcel against the mat stops with the closed gripper is not a violation.

SOP rule broken: Steps 1 to 5, a bad placement is fixed by lifting the object clear and placing it again, and each check allows only the fix it states.

Coaching note: nothing on this station is pushed into line except the parcel onto its two stops. If the same fix keeps coming back, let review flag it and move on.

Violation: Carry over a working spot

Visible cue: the parcel, the bar or a sticker is carried over the log card, over the scale after the parcel has come off the pan, or over a sticker pad it is not going to; or a carry crosses the far side of the table.

SOP rule broken: Workspace layout, nothing is carried over the card or over the scale, and nothing travels to the far side of the table.

Coaching note: go around, not across. Move back or forward on the line first, then straight across.

Violation: Dropped, collided, or knocked over

Visible cue: under command, the gripper or a carried object hits the scale, the mat stops, the seated parcel, the card, the pen holder or a sticker pad and knocks, tips, slides or moves it; or an object falls short of its spot, with no gripper failure, drift, or controller fault visible.

SOP rule broken: Steps 1 to 5, keep each object on a clear path and let go only after a settled placement.

Coaching note: watch the whole path, not just the target. A knock on the scale or the mat spoils every number after it.

Violation: Work outside the camera frame

Visible cue: a weighing, a reading off the bar, a writing, a mark or a sticker placement happens partly or wholly outside the environment camera frame, or the display, the bar marks or the card are blocked while a number is being read or written.

SOP rule broken: Steps 1 to 5, every move happens inside the camera frame and every number is readable while it is used.

Coaching note: the work has drifted outside the framed area. Set the spots back to their setup places and keep the gripper off the numbers.

Violation: Wrong episode ending

Visible cue: the episode ends with the parcel still on the pan or off its stops, a number box blank, no band box marked, no sticker on the parcel, a tool off its spot, short of home, or the gripper not fully open.

SOP rule broken: Step 6, check the parcel, the card, the display and the tools, then go home with the gripper open and stop recording.

Coaching note: check first. Going home is the last thing that happens before the recording ends.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, throw the episode away, and never use them for coaching.

Recording stopped or paused during the episode (recording system).

Camera dropped frames or lost its feed (capture system).

Hardware fault: gripper failure, drift, controller caused collision, or motor error.

A spot sits outside comfortable reach, so the move stalls at a joint limit or has to fold in on itself to work the spot. Move the spot and run the Setup checklists again.

Bad scale: one that will not hold zero, that drifts under a settled parcel, that goes blank or drops the number when the parcel is lifted off, or whose display cannot be read on camera.

Rocking table: the display moves off zero when the tabletop is pressed beside the scale. Steady the table or move the scale before the next episode.

Bad mat: one that slides or lifts under a correct seating, or whose stops are not square. Weigh it down, clamp it or replace it before the next episode.

Bad bar: one that is bent, whose marks are rubbed out or unreadable, or whose end stop is loose.

Bad parcel: one that squashes under the gripper, whose top face bows, or that is open, crushed or wet, so no reading and no sticker can sit flat on it.

Bad sticker from a pad: one that will not come off the pad, has lost its stick, or carries a smeared or blank word. Replace the pad before the next episode.

Dry pen at episode start, or one that will not write on a clean card.

In spot holding no parcel, or a pad holding no sticker of the band the chart gave. Reset error, not a violation.

Annotation subtasks (from SOP)

1. Press the ZERO bar on the scale
2. Take one parcel from the in spot
3. Set the parcel on the scale pan
4. Read the weight off the display

5. Move the parcel onto the measure mat
6. Seat the parcel square against both stops
7. Take the bar from its spot
8. Measure the length with the bar
9. Measure the width with the bar
10. Measure the height with the bar
11. Return the bar to its spot
12. Take the pen from its holder
13. Write one number on the card
14. Return the pen to its holder
15. Mark one band box on the card
16. Peel one sticker from its pad
17. Lay the sticker on the top face
18. Press the sticker flat
19. Go home and end the episode